Naeed Akhtar

Agentic & Robotics AI Trainer

Mathematics · Physics · Computer Science | Three master's degrees

50 papers

PROFILE

Mathematics, Physics and Computer Science graduate with **three master's degrees**: Mathematics (University of Rajasthan, CGPA 9.5), Computer Science (University of Delhi, CGPA 8.9) and Physics (Jamia Millia Islamia, CGPA 9.7). I am looking for work as an **Agentic & Robotics AI Trainer**.

AI training needs people who can check an AI's reasoning step by step, judge whether its plans and code are correct, and tell whether a robot's actions make physical sense. These are the three skills my degrees train: rigorous mathematical reasoning, an understanding of the physical world, and algorithms, search and planning. I know five languages: English and Hindi (native), Urdu (professional working), Persian (fluent) and Arabic (basic).

HOW MY BACKGROUND FITS AI TRAINING WORK

Agentic AI training

AI systems that plan and carry out a task in several steps

- Review the plan, the reasoning behind each step and the final result
- Find the first wrong step in a long chain of reasoning
- Check code, algorithms and the logic of search, planning and decision rules
- Master's coursework: intelligent agents, search and game playing, knowledge representation, planning, computational intelligence

Robotics AI training

AI for robots and physical systems, where physics decides what is correct

- Judge whether an action, or a sequence of actions, is physically possible
- Check cause and effect with mechanics, energy, electromagnetism and thermodynamics
- Use units, conservation laws and sanity checks to catch impossible results
- Master's coursework: classical mechanics (Lagrangian and Hamiltonian forms, rigid bodies, nonlinear dynamics), electrodynamics, numerical analysis

EDUCATION

2023 – 2025	**Master's in Physics** · Jamia Millia Islamia, New Delhi Quantum and classical mechanics, mathematical physics, electrodynamics, statistical mechanics, condensed matter, numerical analysis and programming	**CGPA 9.7**
2020 – 2022	**Master's in Computer Science** · University of Delhi Design and analysis of algorithms, artificial intelligence (agents, search, planning), computational intelligence, data mining, databases, operating systems, networks	**CGPA 8.9**
2016 – 2018	**Master's in Mathematics** · University of Rajasthan, Jaipur Algebra, real analysis, topology, functional analysis, differential equations, differential geometry, calculus of variations, integral transforms	**CGPA 9.5**
2011 – 2014	**Bachelor's in Mathematics, Physics and Computer Science** · University of Delhi Calculus, algebra, real analysis, differential equations; mechanics, thermal physics, waves and optics, electromagnetism, quantum physics; C++, data structures, operating systems, networks, databases	**CGPA 9.6**
2008 – 2010	**Classes XI–XII, Science** · Mayoor School Mathematics, Physics and Computer Science (C++)	**87.50%** First Division

University results are CGPA out of 10; Classes XI–XII are given as a percentage with the division. Course topics follow the published syllabus of each programme.

Research papers reviewed: 50. Full list with a link to every paper: portfolio.naeed.workers.dev/papers

CORE SKILLS: THREE SUBJECTS, ONE PROFILE

Mathematics

In AI training: follow a multi-step solution line by line, find the first wrong step, and judge whether a numerical answer is accurate and stable.

- Algebra: groups, rings, fields, Galois theory
- Real analysis and measure theory
- Topology and functional analysis
- Ordinary and partial differential equations
- Differential geometry and tensor analysis
- Calculus of variations
- Integral transforms and integral equations

Physics

In AI training: tell whether a robot's action is physically possible and whether a chain of cause and effect obeys mechanics, energy and electromagnetism.

- Classical mechanics: Lagrangian and Hamiltonian forms, rigid bodies, chaos
- Quantum mechanics
- Electrodynamics
- Statistical mechanics
- Condensed matter physics
- Mathematical physics: complex analysis, PDEs, tensors
- Numerical analysis and programming

Computer Science

In AI training: trace code, algorithms and plans step by step, and know how search, planning and intelligent agents work.

- Design and analysis of algorithms, graph algorithms, NP-completeness
- Artificial intelligence: agents, search, planning, knowledge representation
- Computational intelligence: fuzzy logic, neural networks, evolutionary computation
- Data mining, databases
- Operating systems, computer networks, information security
- C++ programming

TASKS I CAN SUPPORT

- Reviewing and correcting multi-step mathematics and science solutions
- Checking code and the explanation of an algorithm
- Evaluating the plan of an AI agent: steps, order and consistency
- Checking robot action sequences against physics and cause and effect
- Explaining in writing why an answer is right or wrong
- Working in English and Hindi (native), Urdu (professional working) and Persian (fluent)

RESEARCH INTEREST

PROPOSED RESEARCH DIRECTION

Computational and Mathematical Modelling of Complex Physical Systems

The aim is to describe complex physical systems with mathematical models, check those models against the principles of physics, and solve them with algorithms and simulations. It uses all three subjects, and it is also the way I look at agentic and robotic AI: equations for the reasoning, physics for the real world, algorithms for the plan.

Model

Mathematics: describe physical systems with mathematical equations.

FOCUS AREAS

- Differential equations
- Nonlinear dynamics
- Stability

Validate

Physics: check the models against the principles of physics.

FOCUS AREAS

- Conservation laws
- Symmetries
- Dimensional analysis

Solve

Computer Science: solve them with algorithms and simulations.

FOCUS AREAS

- Numerical methods
- Simulation
- Scientific computing

LANGUAGES

English native Hindi (हिन्दी) native Urdu (اردو) professional working Persian (فارسی) fluent Arabic (العربية) basic

Open to Agentic & Robotics AI Trainer roles. The portfolio shows the full education with topics and subtopics for each subject, and the list of research papers reviewed. Contact: **akhtarnaeed1984@gmail.com**